Retention Strategy Memo - Online Retail II

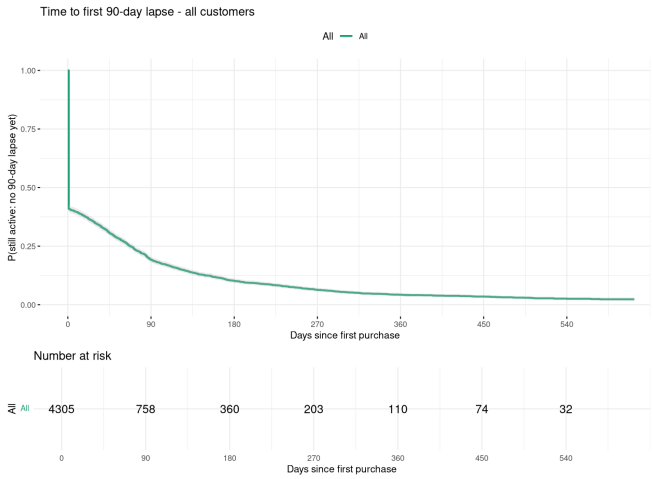
To: Commercial leadership | **From:** Customer Analytics | **Scoring date:** 2011-09-01 (back-test) **Basis:** UCI Online Retail II, 2009-12-01 to 2011-12-09; every figure below is computed by the pipeline in this repository. Items marked **ASSUMED** are scenario inputs, not observed results.

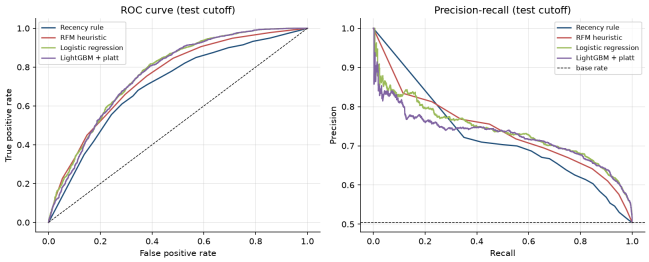
1. The decision

Which active customers should receive retention spend in the next quarter, and how much is their future purchasing plausibly worth?

2. What the data says (observed)

- **5,852 identifiable customers** placed 36,592 clean invoices worth GBP 16.82m. 28% bought only once.
- **Activation is the biggest leak.** Only 41% of new customers made a repeat purchase within 90 days of their first order; 4.1% were still active without any 90-day lapse after a year (Kaplan-Meier). Customers whose first order was in the top value quartile stayed active at 90 days at 26% vs 12% for the bottom quartile; each 1-unit increase in log first-order value cuts the lapse hazard by 12% (stratified Cox HR 0.88, 95% CI 0.84-0.92).
- **Churn proxy** (no purchase in the 90 days after the scoring date - a behavioural proxy, not a cancellation): 50.5% of the 4,323 customers active in the prior year at 2011-09-01.





3. How well can we predict it (out-of-time test, 2011-09-01)

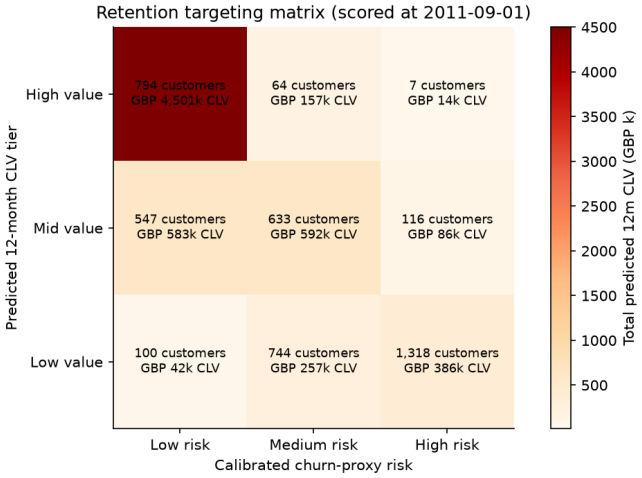
Approach	ROC-AUC	PR-AUC	Lift, top 10%
Recency rule (baseline)	0.706	0.670	1.45x
RFM heuristic (baseline)	0.750	0.715	1.64x
Logistic regression + platt (selected on validation)	**0.769**	**0.737**	**1.64x**

The model ranks risk better than the baselines, but the gain over a simple RFM score is modest (+0.019 ROC-AUC). **Calibration caveat:** at this pre-Christmas scoring date the model predicted an average churn risk of 63% versus 50% observed, consistent with seasonal Q4 demand lowering churn (training windows mix seasons). Use the scores to *rank* customers; re-calibrate on the latest season before using them as absolute probabilities.

Value forecast (BG/NBD + Gamma-Gamma): on a 191-day holdout the model predicted 8,003 repeat purchases vs 8,065 actual and GBP 3.97m spend vs GBP 4.52m actual (spend under-forecast by 12%; the holdout contains the Q4 peak and the model has no seasonality). The top 10% of customers by predicted value delivered 64% of actual holdout spend.

4. Value x risk segments at 2011-09-01

- **Protect - High value / Low risk:** 794 customers, GBP 4.50m predicted 12-month CLV (68% of the total). Observed next-90-day churn was only 12% and they generated 68% of next-quarter revenue. Service, not discounts.
- **Save - High value at Medium/High risk:** 71 customers, GBP 172k CLV; observed churn 48%. Highest priority for personal outreach.
- **Grow - Mid value at Medium/High risk:** 749 customers, GBP 677k CLV (Mid/Medium alone: 633). Low-cost automated win-back.
- **Low value / High risk:** 1,318 customers, mean CLV GBP 293: email-only or no spend.



5. Recommended targeting rule (scenario - ASSUMED inputs)

Contact a customer when *P(churn) x uplift x CLV(12m) x margin > contact cost*. With **ASSUMED** 10% relative churn reduction, 30% margin and GBP 5 per contact, the rule selects 3,657 customers (85%), costing GBP 18k with an expected net value of GBP 35k. Under a conservative scenario (5% uplift, GBP 10 contact) it selects 690 customers for GBP 3,906. Across the 48-scenario grid the expected net value ranges from GBP 326 to GBP 219k. **These are not observed impacts**; the probabilities feeding them are over-stated at this date (section 3).

6. Next steps

1. Run a randomised hold-out test (e.g. 20% control) in the Save and Grow cells to *measure* uplift; replace the assumption with the measured value.
2. Launch a second-order activation programme for new customers (the day-1 lapse is the largest loss).
3. Re-score monthly with `churn score --cutoff <date>` and re-calibrate on the most recent comparable season.

Limitations: wholesale-heavy, 91% UK customer base; 2 years of history (one Q4 per split); 243,007 raw lines lacked a customer ID and are excluded; churn is a proxy.